Shivam Maurya

Data Scientist | Machine Learning, NLP, LLM Applications and Explainable AI

Pune, India | +91 7009411956 | shivammaurya142005@gmail.com

linkedin.com/in/shivammaurya14 | github.com/ShivamMaurya14 | my-portfolio-nu-amber-83.vercel.app

SUMMARY

Data Scientist (B.E., CGPA 8.06 / 10) with hands-on experience across the full machine learning lifecycle: data pipelines, feature engineering, modeling, evaluation, explainability, and deployment, plus applied NLP and LLM / RAG systems. Built a credit-risk model on 307,511 records (ROC-AUC 0.8866, 91.46% default recall) with SHAP reason codes served through FastAPI, and benchmarked 7 classifiers to reach 99.27% accuracy (Top Submission, CODECURE 2025, IIT BHU). Also built LLM-powered agents with retrieval-augmented generation that reduced hallucinations by 85%. Amazon ML Challenge 2026 Top 500 (All India); 15+ merged open-source pull requests.

TECHNICAL SKILLS

Programming and Data: Python (Pandas, NumPy, scikit-learn), SQL, C++, Bash, Data Pipelines, Data Cleaning, Feature Engineering, Large Dataset Analysis, SQLite

Machine Learning: Supervised Learning, Classification, Regression, Random Forest, XGBoost, LightGBM, Ensemble Methods, Cross-Validation, Hyperparameter Tuning, Threshold Optimization, Model Calibration, Model Benchmarking

Model Evaluation and Explainability: ROC-AUC, Precision, Recall, F-score, Confusion Matrix, SHAP, LIME, Grad-CAM, Explainable AI (XAI)

Deep Learning: PyTorch, TensorFlow, Keras, CNN, Transfer Learning (Xception), ONNX Runtime, Computer Vision, OpenCV

NLP and Generative AI: Natural Language Processing (NLP), Large Language Models (Gemini, Llama 3.3, Gemma), Retrieval-Augmented Generation (RAG), Vector Databases (Pinecone), LangChain, LangGraph, Prompt Engineering, AI Agents, OCR (Tesseract), Document Parsing (LlamaParse, PyMuPDF), Intent Classification

MLOps and Deployment: FastAPI, Flask, REST APIs, Docker, Git, GitHub, CI/CD, Model Serving, Streamlit Dashboards, Linux

Foundations: Probability and Statistics, Data Structures and Algorithms, Database Management Systems, Experiment Design

EXPERIENCE

Physical AI Intern | Nextup Robotics | Remote | Aug 2026 - Oct 2026

- Built a synchronized multimodal data-generation pipeline (20 RTX cameras plus tactile contact sensors across 7 rigid bodies per robot) producing vision and tactile datasets for NVIDIA Cosmos foundation models.
- Scaled a simulation to 30 UR10e robots and 20 cameras at 40-50 FPS on a single consumer GPU workstation through profiling and buffer tuning.
- Authored 7 engineering guides covering setup, calibration, and headless CI deployment.

Software Lead | Centre of Excellence for AI & Robotics, Army Institute of Technology, Pune | Jun 2024 - Present

- Trained and validated 4+ ML architectures (CNN, Xception transfer learning, Random Forest, XGBoost) using stratified cross-validation, data augmentation, and automated hyperparameter sweeps.
- Built an ONNX export pipeline and a JSON-configurable inference backend that enables runtime model swaps and A/B testing with zero downstream code changes, reaching 30+ FPS on CPU-only hardware.
- Led software for the ABU Robocon national robot (All India Rank under 20) and directed a 100+ member research cell across 5+ teams.

Open-Source Contributor | ROS 2 Ecosystem (navigation2, ros2_control, ros2_controllers) | Jun 2025 - Present

- Authored 15+ merged pull requests; diagnosed a controller crash from backtraces, fixed cross-repository CI failures, and added static type checking to improve reliability.

PROJECTS

Credit Risk Intelligence System | Python, scikit-learn, Random Forest, XGBoost, SHAP, FastAPI | github.com/ShivamMaurya14/CreditRisk-Predictor

- Built an end-to-end pipeline on 307,511 loan applications with 125 engineered features, achieving ROC-AUC 0.8866 and 91.46% default recall.
- Optimized the decision threshold for net financial benefit rather than raw accuracy, and kept research and production in sync through a shared calibration file.
- Served predictions with SHAP reason codes through a FastAPI service so every decision is explainable.

Antibiotic Resistance Predictor | scikit-learn, XGBoost | Top Submission, CODECURE 2025, IIT BHU | github.com/ShivamMaurya14/Antibiotic-Resistance-Predictor

- Benchmarked 7 classifiers to predict imipenem resistance from cheaper antibiotic susceptibility tests, reaching 99.27% accuracy.

MedSynapse: Clinical AI Platform with 9 ML and Deep Learning Modules | FastAPI, React, PyTorch, TensorFlow, Groq, Ollama | IIIT Pune Finalist, Avishkar 2026 | github.com/ShivamMaurya14/MedSynapse

- Built 9 diagnostic models (up to 99.25% accuracy) with an LLM layer that synthesizes results into calibrated reports with hallucination safeguards.
- Processed unstructured clinical data (text, audio, X-rays, MRI, fundus images, lab PDFs) with Tesseract OCR and PyMuPDF, stored in a SQLite feature store.
- Added SHAP, LIME, and Grad-CAM explainability with SHAP-LIME agreement scoring for confidence-aware outputs.

CREDI-MITRA: AI Credit Appraisal Agent | LangGraph, RAG, Pinecone, XGBoost, Gemini, Groq | Hackathon 2026 Runner-Up | github.com/ShivamMaurya14/CREDI-MITRA

- Built a RAG pipeline with LlamaParse and Pinecone over unstructured financial PDFs that reduced hallucinations by 85%.
- Deployed an XGBoost model (97% accuracy) predicting loan approval, credit limit, and interest rate inside a 5-phase LangGraph agent with human-in-the-loop review.

Industrial IoT Predictive Maintenance Platform | FastAPI, Streamlit, Random Forest, ESP32 | github.com/ShivamMaurya14/Predictive-Maintenance-System | Live demo

- Built a decoupled system with a FastAPI ingestion API for live sensor data and a Streamlit dashboard, predicting equipment failure with over 98% accuracy.

Edge Voice Activator for Home Automation | Python, TFLite, Vosk ASR | Smart India Hackathon 2026 (ISRO PS-26172)

- Designed a 3-stage NLP intent parser (Regex, Fuzzy Levenshtein matching, ML fallback) reaching over 97% intent accuracy on noisy speech-recognition output.

EDUCATION

Bachelor of Engineering (B.E.), Automation and Robotics Engineering | Army Institute of Technology, Pune, India | Jun 2024 - May 2028 (Expected) | **CGPA: 8.06 / 10**

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Probability and Statistics, Data Structures and Algorithms, Database Management Systems (SQL), Object-Oriented Programming

HONORS & ACHIEVEMENTS

- Top 500 All India, Amazon ML Challenge 2026 (F0.5 score 0.9602)
- Top Submission, CODECURE Hackathon 2025, IIT BHU (99.27% accuracy)
- Runner-Up, Hackathon 2026 (CREDI-MITRA)
- State Level Round 2, Avishkar Maharashtra Inter-University Research Convention 2026 (MedSynapse)
- Finalist, IIIT Pune Thinking Machines 2025 (MedSynapse)
- All India Rank under 20, ABU Robocon 2025/26

Languages: English (Professional Working Proficiency), Hindi (Native)